The four-colour Rado number of $2(x-y)=3z$ is 226

A cube-and-conquer computation with checked DRAT proofs

Takumi Ichikawa
github.com/IchiTaku326

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Abstract

For a linear equation E and $k \geq 1$, the Rado number $\mathrm{R}_k(E)$ is the least n such that every k-colouring of $\{1,\dots,n\}$ contains a monochromatic solution of E. Chang, De Loera and Wesley [1] computed many such numbers by SAT solving; for the equation $2(x-y)=3z$ and $k=4$ colours they report only the lower bound $\mathrm{R}_4 > 225$. We show that $\mathrm{R}_4(2(x-y)=3z)=226$. The lower bound is an explicit 4-colouring of $\{1,\dots,225\}$, checked by an independent program. The upper bound is a cube-and-conquer proof: 14,162 cubes, each shown unsatisfiable, with every cube re-solved by CaDiCaL and its DRAT proof accepted by `drat-trim`. All programs, logs and the colouring are public.

1 Introduction

Rado's theorem [2] characterises the linear homogeneous equations that are *regular*, i.e. that have a monochromatic solution in every finite colouring of the positive integers. For a regular equation E the Rado number $\mathrm{R}_k(E)$ is finite for every k. For the family $a(x-y)=bz$ with $\gcd(a,b)=1$ the paper [1] proves, among other results, $\mathrm{R}_3(a(x-y)=bz)=a^3$ for $a \geq b+2$, gives the lower bound $\mathrm{R}_k \geq a^k$ for all k, and tabulates computed values for 2, 3 and (for small a,b) 4 colours. In their four-colour table the entry for $a=2$, $b=3$ is ">225"; to our knowledge no exact value has been published (Section 7). This note closes that entry.

Theorem 1. $\mathrm{R}_4(2(x-y)=3z)=226$.

Contributions. The new content is (i) the exact value in Theorem 1, (ii) a complete, resumable pipeline whose every step leaves a time-stamped log, and (iii) a DRAT certificate for each of the 14,162 cases, regenerable in about one hour on a desktop PC.

Conventions. Here a *solution* is a triple (x,y,z) of positive integers with $2(x-y)=3z$; the variables need not be distinct (for example $(4,1,2)$ and $(5,2,2)$ are solutions). Throughout, "colouring of $[n]$" means a map $[n]=\{1,\dots,n\} \to \{0,1,2,3\}$, and it is *good* if it has no monochromatic solution with $x,y,z \leq n$. Then $\mathrm{R}_4 > n$ iff a good colouring of $[n]$ exists, and Theorem 1 is the conjunction of the two statements below.

Theorem 2 (Lower bound)**.** *There is a good* 4*-colouring of* [225].

Theorem 3 (Upper bound)**.** *There is no good* 4*-colouring of* [226].

2 Lower bound

A good colouring of [225] was found by a SAT solver (CaDiCaL through PySAT) on 20 October 2026 (JST) and is stored in `results/lower_bound_n225/`. It is checked by `tools/rado_verify.py`, a program written to be as plain as possible: three nested loops over $x, y, z \le n$ and a test of $2(x-y) = 3z$ and equality of the three colours; it does not share code with the CNF generator. The check reports `VALID`. A good colouring of [225] restricts to a good colouring of every $[m]$, $m \le 225$, so $\mathrm{R}_4 > m$ for all $m \le 225$.

3 Upper bound

3.1 Encoding

For $n = 226$, $k = 4$ we use Boolean variables $v_{j,c}$ ($j \in [n]$, $c \in \{0,1,2,3\}$): "j has colour c". The clauses are:

1. each j gets at least one colour, and at most one colour (6 binary clauses per j);
2. for every *solution set* $S = \{x, y, z\}$ and every colour c the clause $\bigvee_{j \in S} \neg v_{j,c}$ (the set S may have two elements if $z = x$ or $z = y$); there are 8,400 solution sets in [226];
3. symmetry breaking (colours numbered in order of first use): auxiliary variables $u_{j,c}$ meaning "colour c occurs in $1..j$" ($c = 0, 1, 2$) with the defining clauses, together with $v_{1,c} \to \bot$ for $c \ge 1$ and $v_{j,c} \to u_{j-1,c-1}$ for $c \ge 1$.

The formula has 1,582 variables and 37,891 clauses. Every good colouring can be renamed so that colours first appear in the order $0, 1, 2, 3$ (restricted growth); renaming preserves goodness, and the renamed colouring satisfies the formula. Hence the formula is satisfiable iff a good colouring of [226] exists.

3.2 Cubes

Let $d = 10$. A *cube* is a restricted-growth string of length d over $\{0,1,2,3\}$ (first letter 0; each letter at most one larger than the maximum so far) with no monochromatic solution inside $[d]$. There are exactly 14,162 cubes; this number was obtained by the search program and re-computed by an independent brute-force enumeration of all 4^{10} strings.

Lemma 1. *If a good colouring of* [226] *exists, then the formula together with the unit clauses of some cube is satisfiable.*

Proof. Rename the colours of a good colouring into restricted-growth form. Its restriction to [10] is a restricted-growth string without monochromatic solution, i.e. one of the cubes, and the colouring satisfies the formula by the previous subsection. □

3.3 Cube search

Each of the 14,162 cubes was solved by Glucose (via PySAT, assumptions, conflict budget $2 \cdot 10^4$, budgets multiplied by 4 on retry) in parallel with 11 workers; all returned UNSAT (21 October 2026, 05:38–09:26 JST, about 41 CPU-hours in total including retried attempts).

3.4 Certification

To avoid trusting an incremental solver, in a second pass (22 October 2026, 01:15–02:23 JST, about 12 CPU-hours) each cube was solved again by CaDiCaL as a separate CNF (formula plus unit clauses), producing a DRAT proof, and `drat-trim` [5] accepted *every* proof: `VERIFIED 14162/14162`. The proofs total about 28 GB (largest: 38 MB) and are therefore not stored; the second pass regenerates them. Lemma 1 and the 14,162 UNSAT certificates give Theorem 3.

4 Computational environment

All computations ran on one desktop PC.

CPU	Intel Core i5-11400F (6 cores, 12 threads)
Memory	32 GB DDR4
GPU	NVIDIA GeForce RTX 3060 (not used; CDCL solvers run on the CPU)
System	Windows; proof checking in WSL (Linux)
Workers	11 processes for the cube search and for the proof check
Lower bound	CaDiCaL 1.5.3 through PySAT
Cube search	Glucose 4 through PySAT, assumptions, budget $2 \cdot 10^4$ conflicts
Proof check	CaDiCaL 3.0.1 (static build) and `drat-trim` (version not recorded)

5 Summary of the computation

Variables / clauses of the formula ($n = 226$)	1,582 / 37,891
Solution sets of $2(x - y) = 3z$ in [226]	8,400
Cubes (depth 10)	14,162
Cubes UNSAT / SAT	14,162 / 0
DRAT proofs accepted by `drat-trim`	14,162 / 14,162
Total size of the proofs	about 28 GB
Cube search, wall clock	3 h 48 min (11 workers)
Proof check, wall clock	about 68 min (11 workers)

6 What this result depends on

We list the places where an error could still hide.

1. **Encoding.** The formula generator is the only place where the problem statement enters the proof. Evidence for its correctness: agreement with exhaustive search on small cases; reproduction of published values (Schur numbers, $\mathrm{R}_4(x - y = 2z) = 171$ and the exact 3-colour values of [1] in the range we tested; 338 comparisons with published values or theorems in a larger table, no disagreement); and validity of the lower-bound colouring under the independent checker.

2. **Symmetry breaking.** The argument above is a pen-and-paper argument; it was not machine-checked.

3. **Cube list.** Completeness rests on Lemma 1 and the count 14,162.

4. **Proof checker.** `drat-trim` is widely used but is itself unverified software; a check with a formally verified checker such as `cake_lpr` [6] has not been done.

5. **Hardware and software.** Runs were done on a single desktop PC (11 worker processes); the version of `drat-trim` was not recorded.

7 Literature status

We checked, on 22 October 2026, the arXiv listing of [1] (only version v1, 7 October 2022, whose four-colour table gives > 225 for $a = 2, b = 3$), a student report from a SAT group on Rado numbers (three colours only), and arXiv:2505.12085 (three colours, other equations); none states a value for this equation. This search was not exhaustive: the list of papers citing [1] and the authors have not yet been consulted. If the value is already known, this note should be regarded as an independent verification.

8 Reproducibility

Code, logs and the colouring: https://github.com/IchiTaku326/rado-226 (MIT licence); archived copy (all versions): https://doi.org/10.5281/zenodo.23565747. Step 0 validates the colouring (seconds), step 1 reruns the cube search (hours on about 11 cores, resumable), step 2 reruns the DRAT check (about one hour on 11 cores). Every row of `results/` carries the JST time at which it was written.

Acknowledgements and disclosure

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References

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